

09.1

Complete **Table 7** with ticks (✓) to show which elements are found in the following biological molecules.

[2 marks]

Table 7

Biological molecules	Element			
	Carbon	Nitrogen	Oxygen	Phosphorus
Galactose				
Phospholipid				
RNA				
Sucrose				

Question 9 continues on the next page

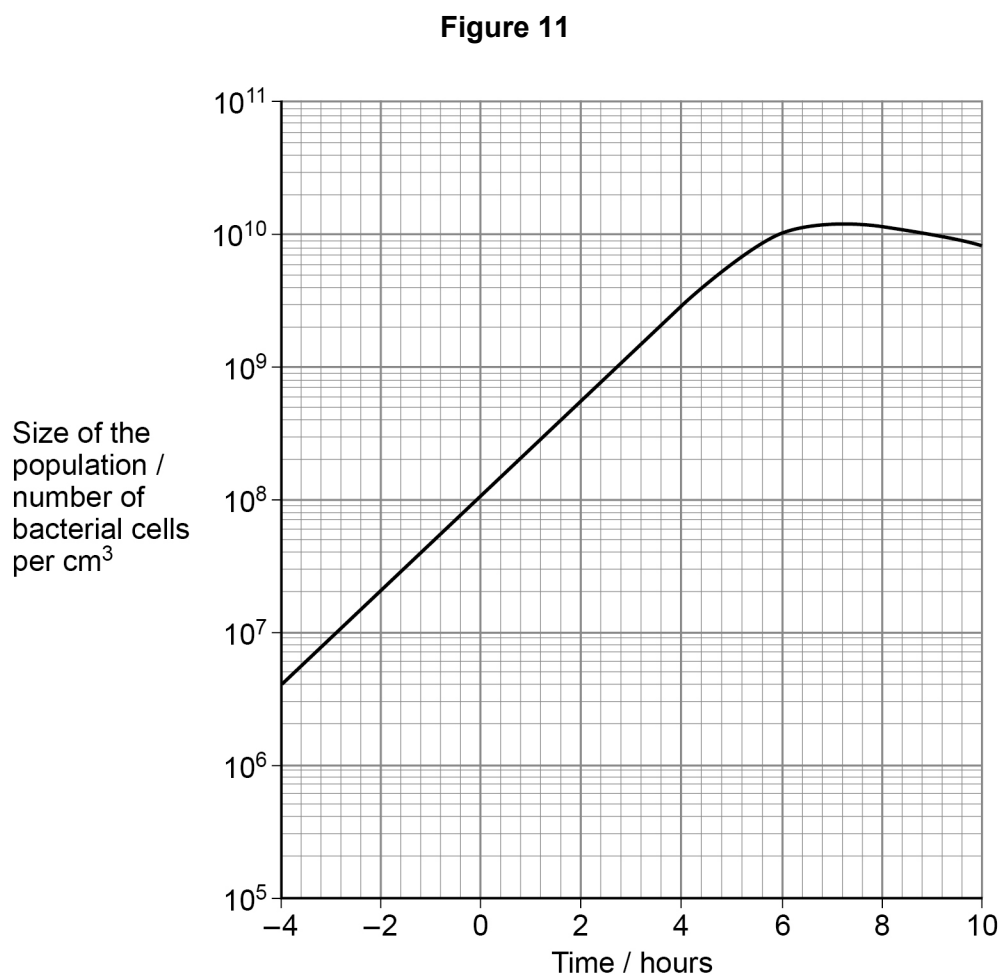
Turn over ►

After Watson and Crick proposed the model of DNA structure, scientists investigated the possible mechanisms for DNA replication.

Two scientists grew a bacterial population, providing them with a nitrogen source containing only the heavy isotope of nitrogen, ^{15}N . As soon as all the DNA in this population contained ^{15}N , the scientists changed the nitrogen source to one containing only the lighter isotope of nitrogen, ^{14}N . They changed the nitrogen source at 0 hours.

During the investigation, the scientists measured the size of the population of bacterial cells.

Figure 11 shows the scientists' results.



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The generation time for a population of bacteria is the time taken for all the bacteria to divide once by binary fission.

Use **Figure 11** and the following equation to calculate the generation time for this population of bacteria. Give your answer in hours.

[2 marks]

$$\text{Number of generations} = \frac{\log_{10} \left(\frac{\text{size of population at time +4 hours}}{\text{size of population at time - 4 hours}} \right)}{\log_{10} 2}$$

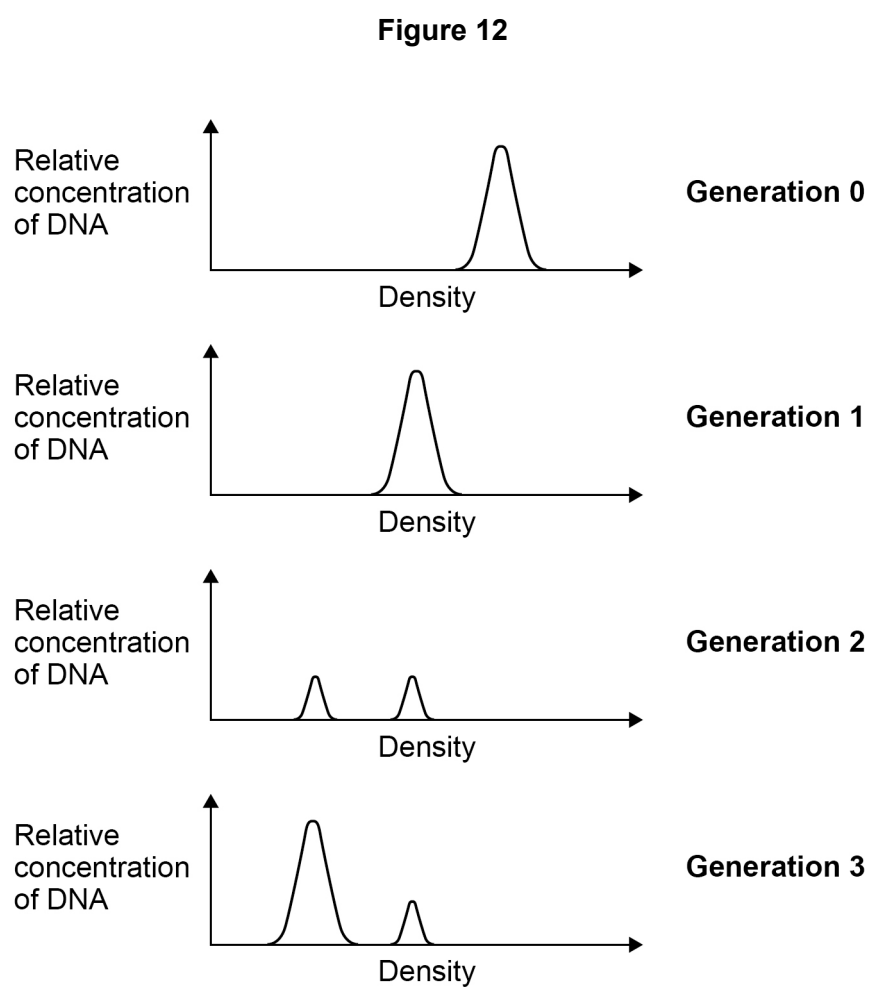
Generation time _____ hours

Question 9 continues on the next page

At intervals during this investigation, the scientists removed samples of the bacterial population, isolated the DNA and measured the density of the DNA.

DNA made using ^{15}N has a higher density than DNA made using ^{14}N .

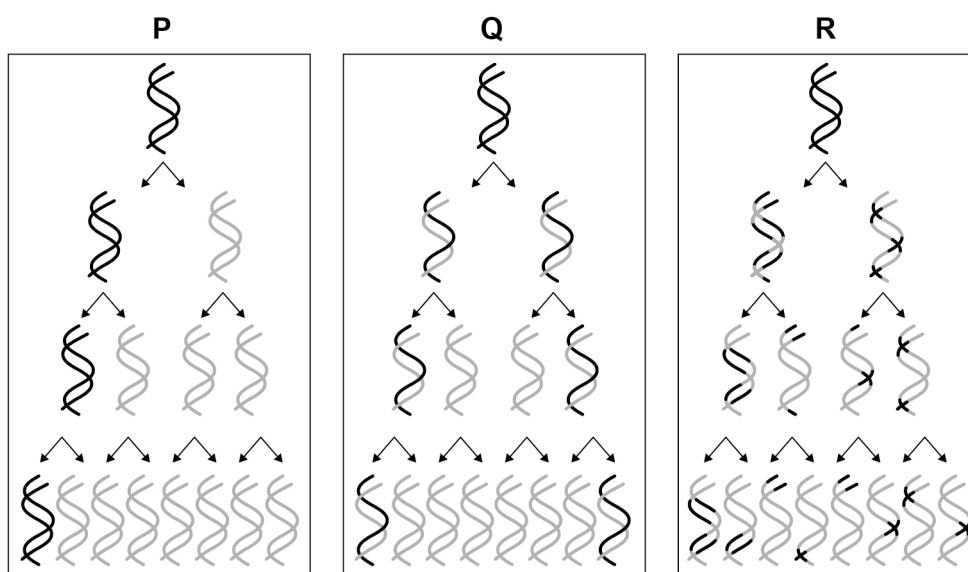
Figure 12 shows the scientists' results.



There are **three** possible models of DNA replication.

These models are shown in **Figure 13**.

Figure 13



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Which of these models, **P**, **Q** or **R**, is supported by the results shown in **Figure 12**?

Give the letter and name of the model supported and explain why the results do **not** support the other models.

[3 marks]

Model _____

Name _____

Explanation for first **unsupported** model _____

Explanation for second **unsupported** model _____
